

The Time Series Low Rank Factor Portfolio

Hema Srikar Ankem

Abstract

This document describes a time series low rank factor portfolio for monthly construction. The model builds a daily panel of managed characteristic portfolios, estimates the low rank structure of that panel on the time series of daily observations rather than on a monthly cross sectional regression, and maps characteristics to exposures over a small number of cross sectionally coupled directions that are combined by a mean variance rule and scaled to a volatility target. We present the managed portfolio construction, the low rank estimation and its ranking criterion, the exposure network, and the overlays.

1 Overview and motivation

Characteristics do not predict returns directly. They set a security's exposure to a small number of common return drivers, which is why the predictable part of the cross section is of low rank. Most models estimate that structure with a monthly cross sectional regression, which uses one observation per security per month. The present model estimates it the other way round, by first turning each characteristic into a managed portfolio and then studying the time series of those portfolios at the daily frequency, so that the covariance of a few hundred portfolios is estimated from thousands of daily observations rather than from a few hundred monthly ones.

2 Characteristic portfolios and their managed daily returns

In each month every characteristic is turned into a long short portfolio by ranking it within the month, demeaning, and scaling to unit gross exposure, so that characteristic j defines a weight vector over the securities. The daily return of that portfolio is obtained by applying the weights formed at the end of month t to the daily returns of the securities during month $t + 1$, so that no information from the future enters. Stacking over months and characteristics yields a daily panel of managed portfolio returns whose columns are the characteristic portfolios and whose rows are trading days.

3 Estimation of the low rank structure

The leading directions of that panel are not chosen by variance alone. Ranking directions by variance rewards how much a direction moves rather than how much it earns, so a direction with a high Sharpe ratio but a low variance would be discarded in favour of a large unrewarded one. To correct this the directions are ranked by the second moment about zero, which is the covariance plus the outer product of the means,

$$M = \widehat{\text{Cov}}(f) + \kappa \bar{f} \bar{f}^\top T, \tag{1}$$

where $\bar{f}$ is the vector of mean managed returns, T is the number of observations, and κ scales the contribution of the mean, with $\kappa = 0$ recovering the usual variance ranking. The leading eigenvectors of M are the estimated directions. Because these directions are individually tradeable combinations of characteristic portfolios, and are not rotation invariant in the way an instrumented factor construction would be, this ranking criterion genuinely changes the portfolio rather than merely relabelling it.

4 The exposure network

A network maps the centred within month ranks of a security’s characteristics to a vector of exposures over a small number of directions equal to the estimated rank, conditioned on a pooled cross sectional context, that is the mean of the security embeddings, so that a security’s exposure reflects the composition of its month. Each exposure column is demeaned and normalised to unit gross exposure, giving a set of cross sectionally coupled component portfolios, and the network is fitted to maximise the realised Sharpe ratio of the equal combination of the components over batches of months. The components are then combined for trading by a mean variance rule estimated from their own realised returns,

$$\lambda = \hat{\Sigma}^{-1} \hat{\mu}, \quad (2)$$

where $\hat{\mu}$ and $\hat{\Sigma}$ are exponentially weighted moments of the component returns, the covariance shrunk towards a scaled identity and lightly ridged. The book before overlays is the combination, normalised to unit gross exposure and averaged across seeds.

5 Overlays and temporal integrity

The component of the book along market beta is projected out without an intercept, so the residual beta is removed while the net exposure survives. The book is then scaled so that its ex ante volatility, measured against a Ledoit and Wolf shrinkage covariance from the trailing daily window, meets a fixed annual target of ten percent, subject to a gross exposure cap. The evidence on this panel records that this overlay contributed more to performance than any change to the signal, which follows from the scale invariance of the Sharpe ratio and locates the gain in the timing of leverage against conditions that the covariance can anticipate. The eigenbasis and the network are both refitted yearly on months up to that year’s cutoff, so that no information from the future enters.

6 Evaluation and concluding remarks

On the evaluation window the model attains a Sharpe ratio of 3.215 in the volatility scaled form and 3.317 in the unscaled form. Estimating the common structure from the daily time series of managed portfolios, rather than from a monthly cross sectional regression, is an effective way to exploit the low rank thesis, since it replaces a small number of monthly observations with a much larger number of daily ones. At the same time the result is consistent with the finding on this panel that compressing the characteristics too far loses information, so the number of retained directions is a working choice that balances the low rank view against the loss of signal that aggressive reduction entails.